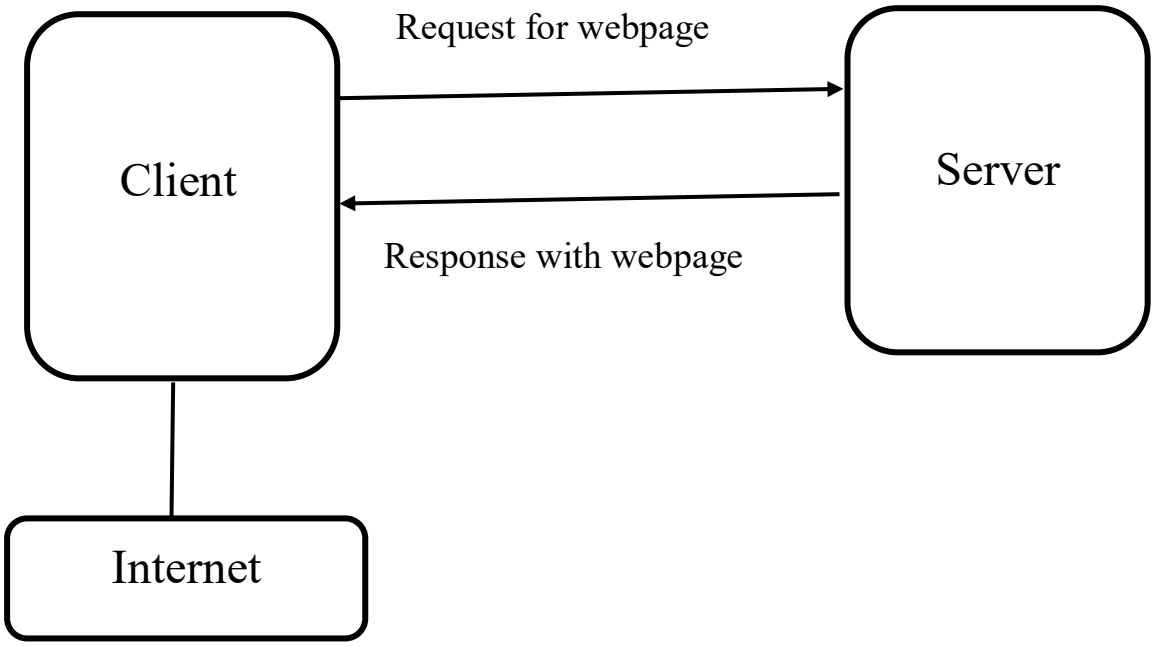
ASSIGNMENT NAME:INTERNET EXPLORER

Client – Server Architecture



Client

- A **client** is the user's device that requests information or services. Examples: Computer, mobile phone, web browser.

- It sends requests to the server.

Server

- A **server** is a powerful computer that stores data, websites, and applications.

  It processes requests from clients and sends back responses.
- It provides services.

Internet

- The **internet** is the network that connects clients and servers worldwide. It allows communication between them using protocols like HTTP/HTTPS.

- It acts as the communication medium.

What Happens When You Open a Website?

Step-by-step process

Step 1: URL Entered

User types website address in browser.

Step 2: DNS Lookup

Browser asks **DNS (Domain Name System)**:

“What is the IP address of this website?”

DNS replies with server IP.

Step 3: Connection Established

Browser creates connection with server using:

- TCP protocol
- Port 80 (HTTP) / 443 (HTTPS)

Step 4: HTTP Request Sent

Browser sends request like:

GET /index.html HTTP/1.1

Step 5: Server Processing

Server:

- Receives request
- Runs application code
- Fetches database data if needed

Step 6: HTTP Response

Server sends back:

- HTML
- CSS
- JavaScript

- Images

Step 7: Browser Rendering

Browser:

- Reads HTML
- Applies CSS
- Runs JavaScript
- Displays webpage

Conclusion

Client-server architecture allows users to access websites by sending requests to servers. When a website is opened, DNS lookup, server communication, and browser rendering happen in milliseconds.